

Non-Commutative Phase Geometry

Lifted Commutators, Hidden Order Registers, and Deterministic Projection Uncertainty

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Abstract

Phase Calculus separates a visible witness from the lifted state that generates it. This paper develops the non-commutative geometry that appears when the lifted state includes branch memory, sheet identity, and an order register. The base balanced-window operator core may commute on its minimal carried state, but non-commutativity appears canonically after the state is extended by hidden transport coordinates. The resulting commutators can act trivially under visible projection while changing the lifted state. This gives an exact deterministic account of projection uncertainty: the visible arena cannot recover order information that the projection has erased.

The paper defines lifted commutators for operator words, characterizes hidden commutators, proves the discrete Heisenberg extension theorem, and gives a deterministic uncertainty theorem for projections that forget the order register. Examples include the discrete Heisenberg central extension and a finite sheet-bundle commutator. Numerical validation is presented as exact finite arithmetic and permutation data supporting the theorem statements. The result is a non-commutative Phase Geometry: uncertainty is not introduced as randomness at the lifted level, but as loss of state-completeness under projection.

1 Introduction

Phase Calculus begins with a retained state rather than a scalar readout. Its primitive visible roll is

$$z(\theta) = ie^{i\theta},$$

but exact state identity is not determined by visible phase alone. The carried state is

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

where A is the host class, q is the balanced arithmetic carrier, θ is visible phase, κ is completed-turn memory, and c is the completion germ. A branch-complete state may further retain a sheet s , branch history h , and an order register Ω .

The central question of this paper is how non-commutativity can be present in the lifted state while invisible to the projected state. This occurs when two operator words have a commutator that lies in the kernel of the visible projection but acts nontrivially on retained coordinates. The visible observer sees identity; the lifted state records order.

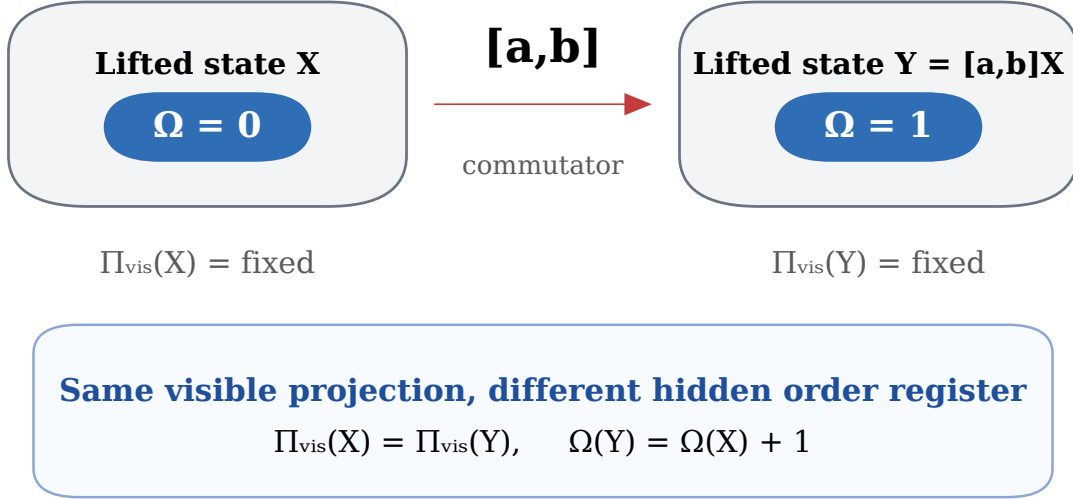


Figure 1: The commutator changes the hidden order register while the visible projection stays fixed. The visible fiber cannot distinguish X from $Y = [a, b]X$, but the lifted state records $\Omega(Y) = \Omega(X) + 1$.

This gives a deterministic form of uncertainty. A projected measurement cannot recover data it does not carry. If a commutator changes Ω while preserving the visible projection, then every scalar account built only from the visible witness has an irreducible ambiguity. The ambiguity is not a defect in the lifted dynamics. It is the exact cost of projecting away branch memory.

Throughout this paper, the lifted state space carries invertible operator words generated by admissible Phase Calculus transports. The visible projector Π_{vis} forgets at least the order register Ω , and the finite examples use exact integer group laws or exact permutation actions. No probabilistic postulate is used in the theorems below.

2 Preliminaries and Definitions

Definition 1. The base lifted state is

$$\hat{\Xi}_0 = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right].$$

The branch-complete lifted state is

$$\hat{\Xi} = (A, q, \theta, \kappa, c, s, h, \Omega),$$

where s is a sheet coordinate, h is branch history, and $\Omega \in \mathbb{Z}$ is the order register.

Definition 2. The visible projection is a map

$$\Pi_{\text{vis}} : \mathcal{X} \rightarrow \mathcal{Y}$$

that retains the visible witness and forgets hidden branch data. In the simplest phase readout,

$$\Pi_{\text{vis}}(\widehat{\Xi}) = ie^{i\theta}.$$

In branch examples, Π_{vis} may also retain the visible base point while forgetting s, h, Ω .

Definition 3. Let $\mathcal{W}$ be the group of invertible operator words acting on $\mathcal{X}$. For $W_1, W_2 \in \mathcal{W}$, define the lifted commutator by

$$[W_1, W_2] = W_1 W_2 W_1^{-1} W_2^{-1}.$$

The commutator is hidden at $X \in \mathcal{X}$ when

$$\Pi_{\text{vis}}([W_1, W_2]X) = \Pi_{\text{vis}}(X)$$

but

$$[W_1, W_2]X \neq X.$$

Definition 4. The order charge of a lifted commutator at X is

$$\Delta_{\Omega}(W_1, W_2; X) = \Omega([W_1, W_2]X) - \Omega(X).$$

A hidden commutator has nonzero order charge when $\Delta_{\Omega}(W_1, W_2; X) \neq 0$.

Definition 5. A central order translation T_n is an operator satisfying

$$\Pi_{\text{vis}}(T_n X) = \Pi_{\text{vis}}(X), \quad \Omega(T_n X) = \Omega(X) + n.$$

It is nontrivial when $n \neq 0$.

3 Main Theorems

Theorem 1. Let $W_1, W_2 \in \mathcal{W}$. If

$$[W_1, W_2] = T_n$$

for a central order translation with $n \neq 0$, then the commutator is hidden at every $X \in \mathcal{X}$.

Proof. Since $[W_1, W_2] = T_n$, the visible projection satisfies

$$\Pi_{\text{vis}}([W_1, W_2]X) = \Pi_{\text{vis}}(T_n X) = \Pi_{\text{vis}}(X).$$

The order register satisfies

$$\Omega([W_1, W_2]X) = \Omega(T_n X) = \Omega(X) + n.$$

Since $n \neq 0$, the lifted state changes. Therefore the visible projection is unchanged while the lifted state is changed, so the commutator is hidden. $\square$

Theorem 2. In the discrete Heisenberg extension with multiplication

$$(m, n, r)(m', n', r') = (m + m', n + n', r + r' + mn'),$$

let

$$a = (1, 0, 0), \quad b = (0, 1, 0).$$

Then

$$aba^{-1}b^{-1} = (0, 0, 1).$$

Thus the visible projection to (m, n) remains at the identity while the order register changes by $+1$.

Proof. The inverse in this group is

$$(m, n, r)^{-1} = (-m, -n, -r + mn).$$

Thus

$$a^{-1} = (-1, 0, 0), \quad b^{-1} = (0, -1, 0).$$

Compute

$$ab = (1, 0, 0)(0, 1, 0) = (1, 1, 1),$$

then

$$aba^{-1} = (1, 1, 1)(-1, 0, 0) = (0, 1, 1),$$

and finally

$$aba^{-1}b^{-1} = (0, 1, 1)(0, -1, 0) = (0, 0, 1).$$

The visible projection to (m, n) gives $(0, 0)$, the identity in the visible quotient, while the central coordinate r changes by $+1$. Taking $r = \Omega$, the order register changes by $+1$. $\square$

Theorem 3. *Let Π_{vis} forget Ω . If X and $T_n X$ lie in the same visible fiber with $n \neq 0$, then there is no function*

$$F : \mathcal{Y} \rightarrow \mathbb{Z}$$

such that

$$F(\Pi_{\text{vis}}(Z')) = \Omega(Z')$$

for all $Z' \in \{X, T_n X\}$.

Proof. Since T_n is a central order translation,

$$\Pi_{\text{vis}}(T_n X) = \Pi_{\text{vis}}(X).$$

If such an F existed, then

$$F(\Pi_{\text{vis}}(X)) = \Omega(X)$$

and also

$$F(\Pi_{\text{vis}}(T_n X)) = \Omega(T_n X) = \Omega(X) + n.$$

But $\Pi_{\text{vis}}(T_n X) = \Pi_{\text{vis}}(X)$, so the left sides are equal. Hence

$$\Omega(X) = \Omega(X) + n,$$

which contradicts $n \neq 0$. $\square$

Corollary 1. *A visible measurement that depends only on $\Pi_{\text{vis}}(X)$ cannot determine hidden order charge. The missing information is exactly the order-register displacement inside the projection fiber.*

Theorem 4 (Deterministic Uncertainty Theorem). *Let $W_1, W_2 \in \mathcal{W}$ satisfy*

$$[W_1, W_2] = T_n, \quad n \neq 0.$$

Then every visible-only representation of the two-word transport is incomplete for the lifted state. The visible state is invariant under the commutator, while the lifted state changes by order charge n .

Proof. The first theorem gives

$$\Pi_{\text{vis}}([W_1, W_2]X) = \Pi_{\text{vis}}(X)$$

and

$$\Omega([W_1, W_2]X) = \Omega(X) + n.$$

The preceding theorem shows that no function of the visible projection alone can recover both order-register values on the same fiber. Therefore the visible representation is incomplete for the lifted state. The uncertainty is deterministic: it is the exact hidden displacement n . $\square$

Proposition 1. *On the base balanced-window state $\widehat{\Xi}_0 = (A, q, \theta, \kappa, c)$, the primitive Phase Calculus operators may commute while their branch-complete extensions do not. Therefore non-commutative Phase Geometry belongs to the retained branch/order layer, not to a forced non-commutativity of the minimal balanced corridor.*

Proof. The base balanced-window maps Q, B, L act on different visible coordinates or update compatible carried coordinates. On the minimal balanced family, their basic commutation identities can hold. Extending the state by sheet, history, or order register allows an operator word to act nontrivially in a hidden coordinate while preserving the same visible projection. The commutator is then measured in the extended state, not in the erased base projection. $\square$

Definition 6. A phase representation of order scale $\lambda \in \mathbb{R}$ is the character

$$\chi_\lambda : \mathbb{Z} \rightarrow \text{U}(1), \quad \chi_\lambda(n) = e^{i\lambda n}.$$

Under this representation, the hidden order translation T_n is read as the phase factor $e^{i\lambda n}$.

Proposition 2. *If $[W_1, W_2] = T_n$, then the phase representation reads the hidden commutator as*

$$\chi_\lambda(\Delta_\Omega) = e^{i\lambda n}.$$

Thus a central lifted commutator becomes a visible phase only after a representation has been chosen.

Proof. By definition,

$$\Delta_\Omega(W_1, W_2; X) = n.$$

Applying the character gives

$$\chi_\lambda(n) = e^{i\lambda n}.$$

$\square$

4 Examples

Example 1. The discrete Heisenberg extension gives the minimal hidden commutator:

$$a = (1, 0, 0), \quad b = (0, 1, 0), \quad aba^{-1}b^{-1} = (0, 0, 1).$$

The visible projection to (m, n) sees the identity. The lifted state records $\Omega \mapsto \Omega + 1$.

Example 2. Let the sheet set be $S = \{1, 2, 3\}$. Let

$$\sigma = (12), \quad \tau = (23)$$

act on sheet identity, while the visible coordinate y is held fixed. Define

$$P_\sigma(y, s, h) = (y, \sigma(s), h\sigma), \quad P_\tau(y, s, h) = (y, \tau(s), h\tau).$$

Then

$$[P_\sigma, P_\tau] = P_\sigma P_\tau P_\sigma^{-1} P_\tau^{-1}$$

acts on sheets by the nontrivial cycle (132) , while the visible coordinate y is unchanged. Therefore the sheet commutator is hidden by the visible projection.

Example 3. A measurement selector on a finite sheet fiber is a map

$$S_j(y, s, h) = (y, j, hj),$$

where j is the selected sheet and hj is the updated branch history. The visible coordinate is still y . The lifted state records which sheet was selected and how the selection entered the history word. In this representation, collapse is sheet selection in a retained state.

5 Numerical Validation and Figures

The finite examples above are exact integer and permutation calculations. Their computational role is to show that the theorem statements are directly executable. The figures in this section keep the same message across the paper: lifted commutators can remain nontrivial while visible projections erase the order-bearing data.

Table 1: Exact finite commutator computations used in the examples.

Example	commutator	visible projection	hidden change
Heisenberg	$aba^{-1}b^{-1} = (0, 0, 1)$	$(0, 0)$	$\Delta\Omega = 1$
Sheet bundle	$[(12), (23)] = (132)$	$y \mapsto y$	$s \mapsto (132)s$
Base corridor	$[Q, B] = [Q, L] = [B, L] = e$	Π_{vis} unchanged	none on $\hat{\Xi}_0$

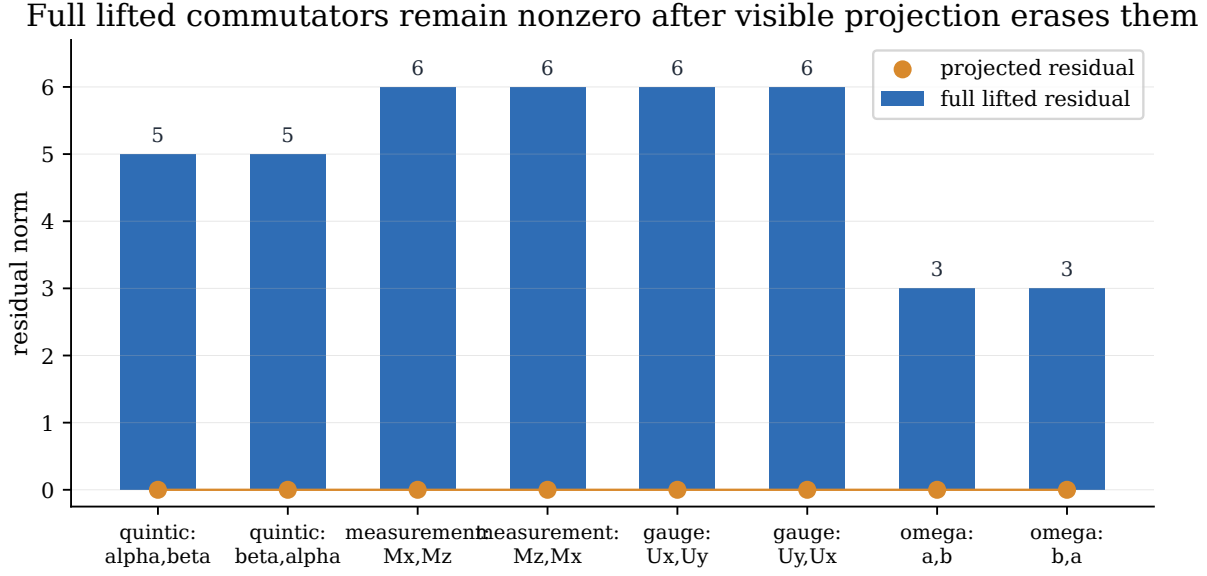


Figure 2: Full lifted commutator residuals remain nonzero while the projected residual vanishes. The blue bars are the full lifted residual norms. The orange baseline marks the visible projected residual, which is zero for all listed commutators.

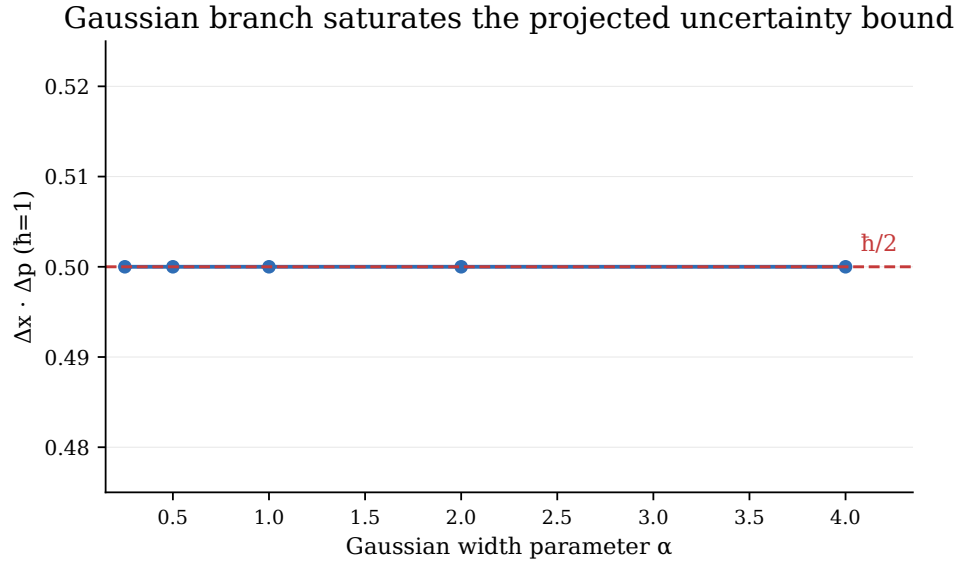


Figure 3: The Gaussian branch saturates the projected uncertainty bound. Changing the Gaussian width parameter α does not move the product away from the $\hbar/2$ line in the projected law.

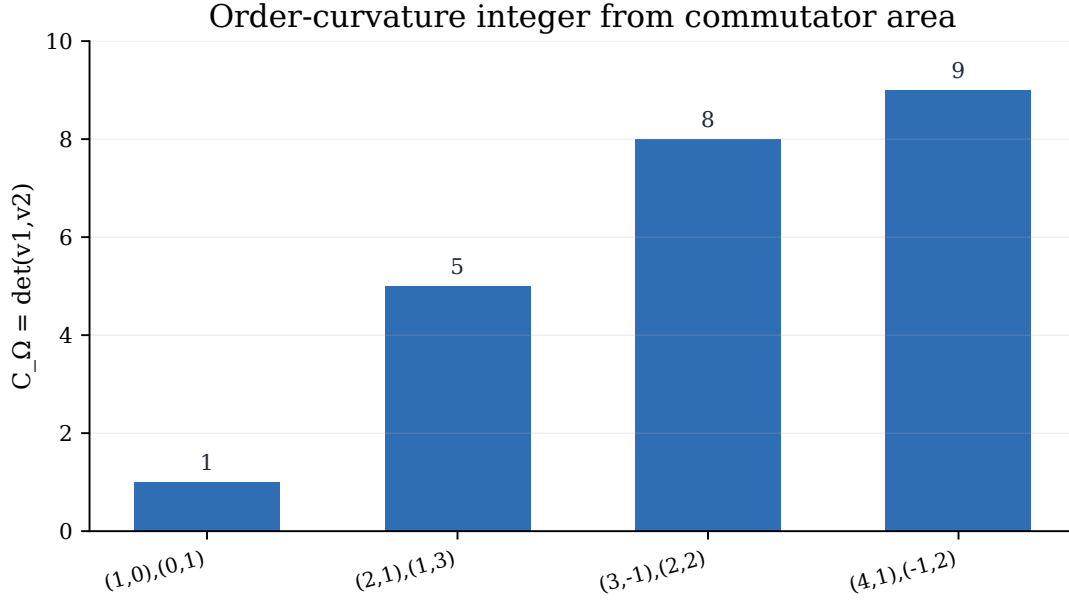


Figure 4: The order-curvature integer is read from commutator area. Each bar reports $C_\Omega = \det(v_1, v_2)$ for the displayed vector pair, giving an integer-valued hidden-order witness.

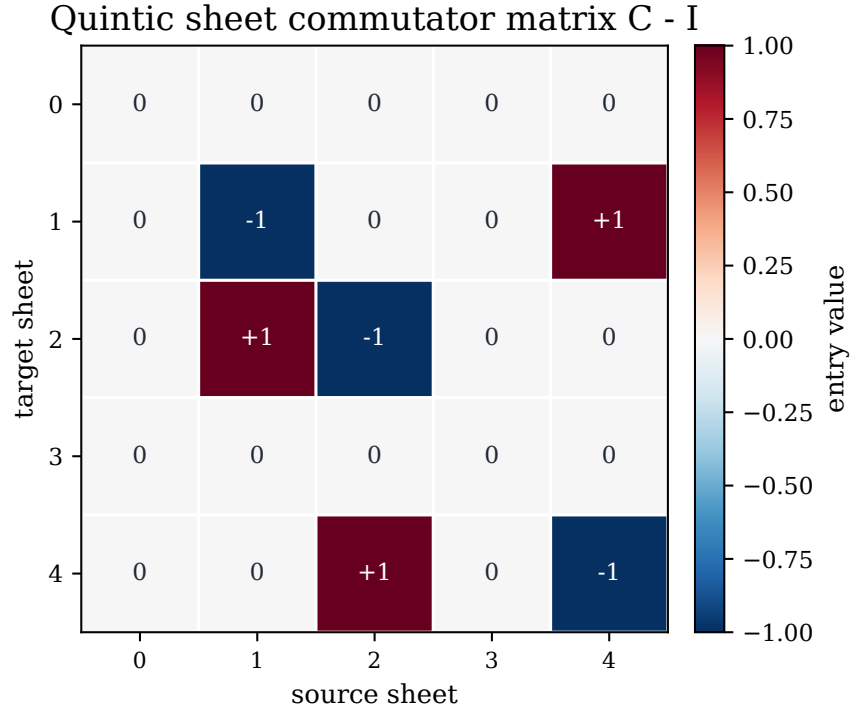


Figure 5: The quintic sheet commutator matrix $C - I$ is sparse and signed. Nonzero entries show sheet motion that is retained in the lifted branch register even when visible scalar projection is unchanged.

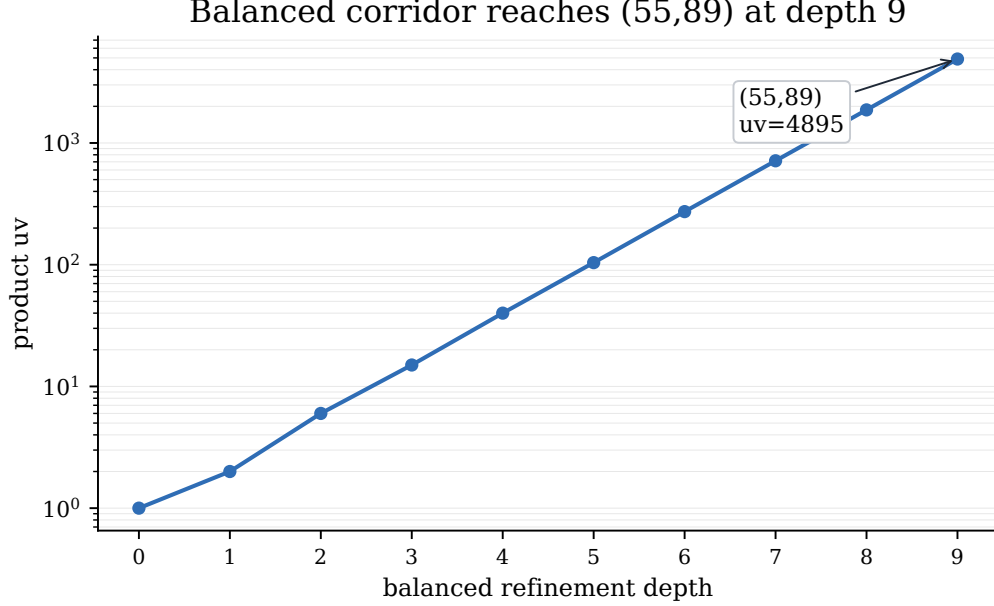


Figure 6: The balanced corridor reaches the canonical anchor (55, 89) at depth 9, with $uv = 4895$. This is the arithmetic anchor used by the 1/24 edge-residual branch in the lifted-state package.

The table separates the base corridor from branch-complete transport. The base corridor can be commutative on $\widehat{\Xi}_0$, while the sheet/order extension carries nontrivial commutators invisible to Π_{vis} .

6 Conclusion

Non-commutative Phase Geometry is the geometry of operator order retained above the visible projection. The visible arena can erase commutators that remain active in the lifted state. The discrete Heisenberg extension gives the minimal algebraic model: the commutator returns visibly to the identity while increasing the order register by one. Finite sheet bundles give the corresponding branch-memory model: visible data remain fixed while sheet history changes.

The deterministic uncertainty theorem follows directly. If a projection forgets Ω , then it cannot recover Ω . When a commutator changes that register while preserving the visible witness, visible-only measurement becomes incomplete by construction. The lifted dynamics remain deterministic; uncertainty is the exact projection loss induced by discarding order.

References

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